

Day 3: SQL Coding Challenge – Clauses & Operators 🛒

Database: ECommerceDB

--1.Create Database

```
CREATE DATABASE ECommerceDB;
```

```
USE ECommerceDB;
```

-- 2.Create Product Table

```
CREATE TABLE Product (  
    product_id INT PRIMARY KEY,  
    product_name VARCHAR(100) NOT NULL,  
    price DECIMAL(10,2) CHECK (price > 0)  
);
```

-- 3.Create Sales Table

```
CREATE TABLE Sales (  
    sale_id INT PRIMARY KEY,  
    product_id INT,  
    quantity INT CHECK (quantity > 0),  
    sale_amount DECIMAL(10,2) CHECK (sale_amount > 0),  
    FOREIGN KEY (product_id) REFERENCES Product(product_id)  
);
```

-- 4.Insert Sample Data into Product Table

```
INSERT INTO Product (product_id, product_name, price) VALUES  
(1, 'Laptop', 85000.00),  
(2, 'Smartphone', 45000.00),
```

(3, 'Headphones', 5000.00),
(4, 'Keyboard', 1200.00),
(5, 'Mouse', 800.00),
(6, 'Monitor', 15000.00),
(7, 'Webcam', 3500.00);

-- 5.Insert Sample Data into Sales Table

```
INSERT INTO Sales (sale_id, product_id, quantity, sale_amount) VALUES  
(1, 1, 2, 170000.00),  
(2, 2, 3, 135000.00),  
(3, 3, 5, 25000.00),  
(4, 4, 10, 12000.00),  
(5, 5, 15, 12000.00),  
(6, 6, 2, 30000.00),  
(7, 7, 4, 14000.00);
```

DISTINCT & AS

1. Display **unique product names** from the Product table.
 2. Display product names with alias **Product_Name**.
 3. Display unique **product_id values** from the Sales table.
 4. Display product price with alias **Product_Price**.
-

WHERE Clause

5. Display products whose **price is greater than 10,000**.
 6. Show products whose **price is less than 5,000**.
 7. Display sales where **quantity equals 2**.
-

Comparison Operators

8. Display products whose **price is greater than or equal to 15,000**.

9. Display sales where **quantity is not equal to 5**.
-

Arithmetic Operators

10. Display product name and **price after 10% increase**.
11. Display sale amount and **sale amount after adding 500**.
-

Logical Operators

12. Display products whose **price is greater than 5,000 AND less than 50,000**.
13. Display sales where **quantity is 2 OR 4**.
14. Display products whose **price is NOT greater than 20,000**.
-

IS NULL / IS NOT NULL

15. Display sales records where **product_id is NULL**.
16. Display products where **price is NOT NULL**.
-

IN and NOT IN

17. Display products with **product_id 1, 3, and 5**.
18. Display products whose **product_id is NOT 2, 4, and 6**.
-

BETWEEN and NOT BETWEEN

19. Display products whose **price is between 1,000 and 20,000**.
20. Display products whose **price is NOT between 5,000 and 50,000**.
-

LIKE and NOT LIKE

21. Display products whose name **starts with 'M'**.
22. Display products whose name **ends with 'e'**.
23. Display products whose name **contains 'phone'**.
24. Display products whose name **does not start with 'S'**.
-

Mixed Questions (Exam Level)

25. Display products whose **price is between 1,000 and 20,000 AND product name starts with 'M'**.
26. Display sales where **quantity is between 2 and 10**.
27. Display products whose **product_id is in (1,2,3) AND price greater than 5,000**.
28. Display unique **sale quantities** from the Sales table.